

Lab Guide 13

Network Policy

Q.1 Create 2 namespaces test and Cassandra and label them with some values

Q.2 create frontend and backend pod using nginx image in test namespace and Cassandra pod in Cassandra namespace.

Q.3 Apply Network Policies

- a. Deny all traffic
- b. allow traffic from frontend to backend in test namespace
- c. from backend to Cassandra pod in Cassandra namespace

Ans: 1

```
kubectl create ns test
```

```
kubectl create ns Cassandra
```

```
kubectl run frontend --image nginx -n test
```

```
kubectl run backend --image nginx -n test
```

```
kubectl run cassandra --image nginx -n Cassandra
```

```
kubectl get pods -n test
```

```
kubectl get pods -n Cassandra
```

```
kubectl -n test exec -it frontend -- curl backend
```

```
kubectl -n test expose pod backend --port 80
```

```
kubectl -n test exec -it frontend -- curl backend
```

```
kubectl -n test exec -it backend -- curl frontend
```

```
kubectl -n test expose pod frontend --port 80
```

```
kubectl -n test exec -it backend -- curl frontend
```

```
kubectl -n test exec -it backend -- curl cassandra.cassandra
```

```
kubectl -n cassandra expose pod cassandra --port 80
```

```
kubectl -n test exec -it backend -- curl cassandra.cassandra
```

Apply deny rules and then allow rules

```
cat 1-default-deny.yml
```

```
kubectl apply -f 1-default-deny.yml
```

```
kubectl -n test exec -it frontend -- curl backend
```

```
kubectl -n test exec -it backend -- curl frontend
```

```
cat 2-frontend_egress.yml
```

```
kubectl get pods -n test --show-labels
```

```
kubectl create -f 2-frontend_egress.yml
```

```
kubectl -n test exec -it frontend -- curl backend
```

```
cat 3-backend_ingress.yml
```

```
kubectl create -f 3-backend_ingress.yml
```

```
kubectl -n test exec -it frontend -- curl backend
```

```
cat 4-default-deny-cassandra.yml
```

```
kubectl create -f 4-default-deny-cassandra.yml
```

```
kubectl -n test exec -it backend -- curl cassandra.cassandra
```

```
cat 5-backend_ingress_cassandra.yml
```

```
kubectl label namespace cassandra ns=Cassandra
```

```
kubectl create -f 5-backend_ingress_cassandra.yml
```

```
kubectl -n test exec -it backend -- curl cassandra.cassandra
```

```
cat 6-cassandra_ingress.yml
```

```
kubectl label namespaces test ns=test
```

```
kubectl create -f 6-cassandra_ingress.yml
```

```
kubectl -n test exec -it backend -- curl cassandra.cassandra
```

```

root@master:~# kubectl create ns test
namespace/test created
root@master:~# kubectl create ns cassandra
namespace/cassandra created
root@master:~# kubectl run frontend --image nginx -n test
pod/frontend created
root@master:~# kubectl run backend --image nginx -n test
pod/backend created
root@master:~# kubectl run cassandra --image nginx -n cassandra
pod/cassandra created
root@master:~# kubectl get pods -n test
NAME          READY   STATUS    RESTARTS   AGE
backend       1/1     Running   0           27s
frontend      1/1     Running   0           31s
root@master:~# kubectl get pods -n cassandra
NAME          READY   STATUS    RESTARTS   AGE
cassandra     1/1     Running   0           21s
root@master:~#

```

```

root@master:~# kubectl -n test exec -it frontend -- curl backend
curl: (6) Could not resolve host: backend
command terminated with exit code 6
root@master:~# kubectl -n test expose pod backend --port 80
service/backend exposed
root@master:~# kubectl -n test exec -it frontend -- curl backend
curl: (6) Could not resolve host: backend
command terminated with exit code 6
root@master:~# kubectl -n test exec -it frontend -- curl backend.test
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@master:~# kubectl -n test exec -it frontend -- curl backend

```

```

root@master:~# kubectl -n test exec -it backend -- curl frontend
curl: (6) Could not resolve host: frontend
command terminated with exit code 6
root@master:~# kubectl -n test expose pod frontend --port 80
service/frontend exposed
root@master:~# kubectl -n test exec -it backend -- curl frontend
<!DOCTYPE html>
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<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@master:~# █

```

```

root@master:~/Kubernetes/main/network-policy# cat 1-default-deny.yml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny
  namespace: test
spec:
  podSelector: {}
  policyTypes:
  - Ingress
  - Egress
  egress:
  - to:
    ports:
    - port: 53
      protocol: TCP
    - port: 53
      protocol: UDP

root@master:~/Kubernetes/main/network-policy# kubectl apply -f 1-default-deny.yml
networkpolicy.networking.k8s.io/default-deny created
root@master:~/Kubernetes/main/network-policy# kubectl -n test exec -it frontend -- curl backend
█

```

```

root@master:~/Kubernetes/main/network-policy# cat 2-frontend_egress.yml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: frontend
  namespace: test
spec:
  podSelector:
    matchLabels:
      run: frontend
  policyTypes:
  - Egress
  egress:
  - to:
    - podSelector:
        matchLabels:
          run: backend
root@master:~/Kubernetes/main/network-policy# kubectl get pods -n test --show-label
error: unknown flag: --show-label
See 'kubectl get --help' for usage.
root@master:~/Kubernetes/main/network-policy# kubectl get pods -n test --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
backend       1/1     Running   0           8m53s   run=backend
frontend      1/1     Running   0           8m57s   run=frontend
root@master:~/Kubernetes/main/network-policy# kubectl create -f 2-frontend_egress.yml
networkpolicy.networking.k8s.io/frontend created
root@master:~/Kubernetes/main/network-policy# kubectl -n test exec -it frontend -- curl backend

```

```

root@master:~/Kubernetes/main/network-policy# cat 3-backend_ingress.yml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: backend
  namespace: test
spec:
  podSelector:
    matchLabels:
      run: backend
  policyTypes:
  - Ingress
  ingress:
  - from:
    - podSelector:
        matchLabels:
          run: frontend
root@master:~/Kubernetes/main/network-policy# kubectl create -f 3-backend_ingress.yml
networkpolicy.networking.k8s.io/backend created
root@master:~/Kubernetes/main/network-policy# kubectl -n test exec -it frontend -- curl backend
<!DOCTYPE html>
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<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

```

```

root@master:~/Kubernetes/main/network-policy# cat 4-default-deny-cassandra.yml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny
  namespace: cassandra
spec:
  podSelector: {}
  policyTypes:
  - Ingress
  - Egress
  egress:
  - to:
    ports:
    - port: 53
      protocol: TCP
    - port: 53
      protocol: UDP

root@master:~/Kubernetes/main/network-policy# kubectl create -f 4-default-deny-cassandra.yml
networkpolicy.networking.k8s.io/default-deny created
root@master:~/Kubernetes/main/network-policy# kubectl -n test exec -it backend -- curl cassandra.cassandra
^Ccommand terminated with exit code 130

```

```

root@master:~/Kubernetes/main/network-policy# cat 5-backend_ingress_cassandra.yml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: backend-ns
  namespace: test
spec:
  podSelector:
    matchLabels:
      run: backend
  policyTypes:
  - Ingress
  - Egress
  ingress:
  - from:
    - podSelector:
        matchLabels:
          run: frontend
  egress:
  - to:
    - namespaceSelector:
        matchLabels:
          ns: cassandra
#   podSelector:
#     matchLabels:
#       ns: backend

root@master:~/Kubernetes/main/network-policy# kubectl label namespace cassandra ns=cassandra
namespace/cassandra labeled
root@master:~/Kubernetes/main/network-policy# kubectl create -f 5-backend_ingress_cassandra.yml
networkpolicy.networking.k8s.io/backend-ns created
root@master:~/Kubernetes/main/network-policy# kubectl -n test exec -it backend -- curl cassandra.cassandra
^Ccommand terminated with exit code 130

```

```
root@master:~/Kubernetes/main/network-policy# cat 6-cassandra_ingress.yml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: cassandra
  namespace: cassandra
spec:
  podSelector:
    matchLabels:
      run: cassandra
  policyTypes:
  - Ingress
  ingress:
  - from:
    - namespaceSelector:
        matchLabels:
          ns: test
      podSelector:
        matchLabels:
          run: backend
#
#
```

root@master:~/Kubernetes/main/network-policy# kubectl label namespaces test ns=test
namespace/test labeled

root@master:~/Kubernetes/main/network-policy# kubectl create -f 6-cassandra_ingress.yml
networkpolicy.networking.k8s.io/cassandra created

root@master:~/Kubernetes/main/network-policy# kubectl -n test exec -it backend -- curl cassandra.cassandra

```
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